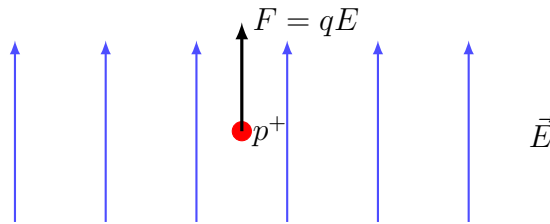


April 27, 2026

1. Determine the acceleration of a proton $m = 1.67 \cdot 10^{-27}$ kg in an electric field of strength $E = 100$ N/C. How many times greater is this acceleration than the acceleration due to gravity?



The electric force acting on the proton is

$$F = qE.$$

Since the proton charge is

$$q = e = 1.602 \cdot 10^{-19} \text{ C},$$

its acceleration is

$$a = \frac{F}{m} = \frac{eE}{m}.$$

Substituting the numerical values,

$$a = \frac{(1.602 \cdot 10^{-19} \text{ C})(100 \text{ N/C})}{1.67 \cdot 10^{-27} \text{ kg}} = 9.59 \cdot 10^9 \text{ m/s}^2.$$

Now compare this with the gravitational acceleration:

$$\frac{a}{g} = \frac{9.59 \cdot 10^9 \text{ m/s}^2}{9.81 \text{ m/s}^2} = 9.78 \cdot 10^8.$$

Therefore,

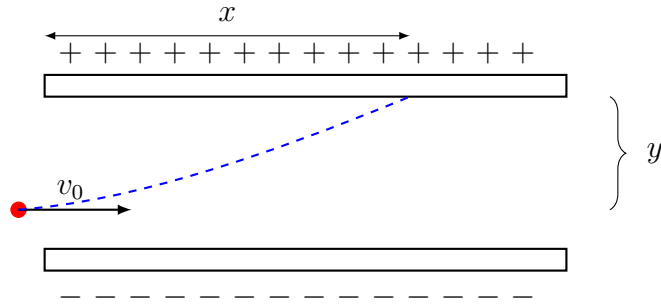
$$a = 9.59 \cdot 10^9 \text{ m/s}^2$$

and the acceleration is approximately

$$9.78 \cdot 10^8$$

times greater than the acceleration due to gravity.

2. An electron is launched with an initial velocity of $v_0 = 2 \cdot 10^6$ m/s between two parallel charged plates. If the electric field strength between the plates is $E = 2$ kN/C, where does the electron hit the upper plate? Assume vacuum.



The magnitude of the electric force is

$$F = eE.$$

Thus the vertical acceleration of the electron is

$$a = \frac{eE}{m_e}.$$

Using

$$m_e = 9.11 \cdot 10^{-31} \text{ kg},$$

we obtain

$$a = \frac{(1.602 \cdot 10^{-19} \text{ C})(2000 \text{ N/C})}{9.11 \cdot 10^{-31} \text{ kg}} = 3.52 \cdot 10^{14} \text{ m/s}^2.$$

The vertical motion is uniformly accelerated:

$$y = \frac{1}{2}at^2.$$

Therefore,

$$t = \sqrt{\frac{2y}{a}}.$$

The horizontal motion is uniform:

$$x = v_0 t.$$

Hence,

$$x = v_0 \sqrt{\frac{2y}{a}}.$$

Substituting the expression for the acceleration:

$$x = v_0 \sqrt{\frac{2m_e y}{eE}}.$$

Now substitute the numerical values:

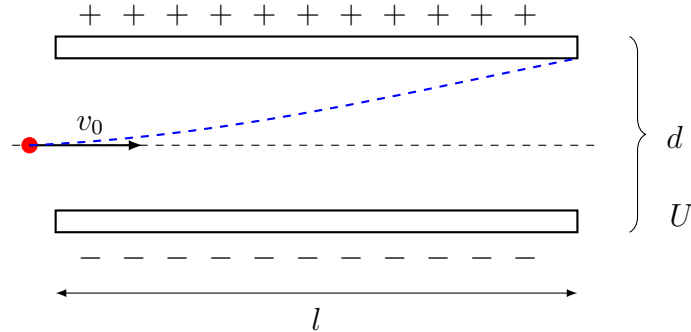
$$x = v_0 \sqrt{\frac{2m_e y}{eE}} = 2 \cdot 10^6 \sqrt{\frac{2(9.11 \cdot 10^{-31})y}{(1.602 \cdot 10^{-19})(2000) \text{ N/C}}}.$$

This gives

$$x = 0.151\sqrt{y},$$

where both x and y are measured in meters.

3. An electron enters the region between the plates halfway between them. The plate length is $l = 10$ cm, their separation is $d = 1$ cm, and the voltage difference is $U = 200$ V. Determine the initial velocity needed for the electron to just barely leave the region between the plates.



The electric field between the plates is

$$E = \frac{U}{d}.$$

Thus,

$$E = \frac{200 \text{ V}}{0.010 \text{ m}} = 2.0 \cdot 10^4 \text{ N/C}.$$

The electron acceleration is

$$a = \frac{eE}{m_e} = \frac{eU}{m_e d}.$$

Substituting the numerical values,

$$a = \frac{(1.602 \cdot 10^{-19} \text{ C})(200 \text{ V})}{(9.11 \cdot 10^{-31} \text{ kg})(0.010 \text{ m})} = 3.52 \cdot 10^{15} \text{ m/s}^2.$$

The electron starts halfway between the plates, therefore its required vertical displacement is

$$\Delta y = \frac{d}{2}.$$

The vertical motion is

$$\frac{d}{2} = \frac{1}{2}at^2.$$

Hence,

$$d = at^2,$$

thus,

$$t = \sqrt{\frac{d}{a}}.$$

The horizontal motion is uniform:

$$l = v_0 t.$$

Therefore,

$$v_0 = \frac{l}{t} = l\sqrt{\frac{a}{d}}.$$

Substituting

$$a = \frac{eU}{m_e d},$$

we get

$$v_0 = l\sqrt{\frac{eU}{m_e d^2}}.$$

Substituting the numerical values,

$$v_0 = l\sqrt{\frac{eU}{m_e d^2}} = 0.10\sqrt{\frac{(1.602 \cdot 10^{-19} \text{ C})(200 \text{ V})}{(9.11 \cdot 10^{-31} \text{ kg})(0.010 \text{ m})^2}} = 5.93 \cdot 10^7 \text{ m/s}.$$

Therefore,

$$\boxed{v_0 = 5.93 \cdot 10^7 \text{ m/s}}$$

is required.

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